RELIABILITY TESTING QUESTIONNAIRE

Apache JMeter Performance and Reliability Evaluation

Integrated Smart Campus Vehicle Management System (ISCVMS)

Camarines Sur Polytechnic Colleges

Filled from the JMeter runs saved on 18 October 2026. Figures are taken from the Summary Report. Where a test was not saved, the status is Not executed rather than marked Pass.

Part A — Tester and Test Environment Profile

Field	Answer
Tester / Respondent Name	Adrian Adams M. Daduya
Role being tested	Researcher
Date of Testing	18 October 2026
Target URL	https://www.iscvms.com
Apache JMeter Version	5.6.3
Machine Specs (CPU / RAM)	AMD Ryzen 5 7520U with Radeon Graphics / 7.2 GB RAM
Network Condition	Other — home internet to the public HTTPS site
Test Plan Name / File	ISCVMS-Reliability.jmx
Modules Targeted	Login, Registration, Portals, RFID API, Reports

The printed form shows http://127.0.0.1:8000. These tests were run against the live site, https://www.iscvms.com.

Part B — Apache JMeter Test Results Log

No.	Scenario	Threads	Ramp-up (s)	Loops / duration	Avg (ms)	Min / Max (ms)	Throughput	Error %	Interpretation
1	Login (normal load) — GET /login	10	10	5 loops (50 samples)	175	72 / 861	5.2/sec	0.00%	Stable. The login page stayed available.
2	Login (moderate load) — GET /login	30	30	3 loops (90 samples)	148	71 / 781	3.1/sec	0.00%	Stable. Faster on average than the 10-user run. No crash.
3	Registration page — GET /register	10	10	3 loops (30 samples)	348	92 / 1571	3.1/sec	0.00%	Page stayed available. Slower than login, still usable.
4	Admin login and GET /admin	1	1	1 (3 samples)	562 on GET /admin	562 / 562	1.8/sec on GET /admin	0.00%	HTTP succeeded. The saved body is the Sign In page, so a logged-in dashboard was not confirmed.
5	RFID API scan — POST	5	10	6 loops (30	186	65 / 1634	3.6/sec	100.00%	Every sample returned

	/api/rfid/scan			samples)					Invalid RFID API token. The API answered and did not crash. The token in this run was rejected.
6	RFID heartbeat — POST /api/rfid/heartbeat	10	10	6 loops (60 samples)	123	47 / 393	6.1/sec	0.00%	Stable. Body was ok true for GATE-IN-1.
7	Mixed portal browsing	5	10	3 loops (115 samples)	171	124 / 470	9.7/sec	0.00%	All labels returned HTTP success. Sampled portal bodies are the Sign In page.
8	Soak / sustained run	10	30	600 seconds (23,189 samples)	251	55 / 2793	38.6/sec	0.00%	Stable for 10 minutes. No errors. Visitor pre-register was the slowest page (avg 502 ms).

Row 4 detail: GET /login 896 ms, POST /login 756 ms, GET /admin 562 ms, all 0.00% error, one sample each.

Row 7 detail, all 0.00% error: GET /login avg 260 ms (5 samples); POST /login avg 169 ms (5); Admin dashboard 179 ms; Registrations 158 ms; RFID assignment 166 ms; Visitors 178 ms; Parking 161 ms; Reports 157 ms; System health 170 ms (15 samples each).

Row 8 detail, all 0.00% error: Home avg 106 ms (5,799); Login page 127 ms (5,799); Register page 269 ms (5,797); Visitor pre-register 502 ms (5,794).

Part C — Reliability Test Cases

No.	Test area	Actual result	Status	Remarks
1	Availability — home/login, normal load	GET /login, 10 users, 50 samples, avg 175 ms, error 0.00%. Login HTML was returned.	Pass	Success rate 100%, above the 98% line.
2	Availability — login, moderate load	GET /login, 30 users, 90 samples, avg 148 ms, error 0.00%. No crash.	Pass	Average was lower than the 10-user run.
3	Reliability — repeated login	Login page stayed at 0.00% error in row 1, row 2, and the 10-minute soak (5,799 login samples, avg 127 ms).	Pass	One POST /login returned HTTP success, but GET /admin then showed the Sign In page.
4	Reliability — registration under concurrent users	GET /register, 10 users, 30 samples, avg 348 ms, error 0.00%. The form HTML loaded.	Pass	A separate invalid POST was not in the saved results. Only two extra GET /register samples were saved (avg 2,462 ms, error 0.00%).

5	Reliability — RFID scan and heartbeat	Scan: 30/30 rejected with Invalid RFID API token, avg 186 ms, site stayed up. Heartbeat after the token was corrected: 60/60, error 0.00%, avg 123 ms, body ok true.	Pass	The scan error was a rejected token, not a crash. Heartbeat confirms the gate API with a valid token.
6	Fault tolerance — invalid passwords	No saved JMeter run of repeated wrong passwords.	Not executed	Do not mark Pass from these files.
7	Fault tolerance — malformed protected request	No saved run with an empty or broken RFID body. The invalid-token scan was rejected cleanly.	Not executed	The token failure is evidence of a clear JSON denial, but it is not the empty-body case.
8	Recoverability — after the stress peak	No separate 10-user rerun was saved after the 30-user test. The 30-user average (148 ms) was already close to, and lower than, the baseline (175 ms). The later soak stayed at 0.00% error.	Pass	There was no slowdown to recover from at 30 users. A 50-user peak was not run.
9	Stability — soak	10 users, 600 seconds, 23,189 samples, avg 251 ms, max 2,793 ms, error 0.00%. JMeter finished the full duration.	Pass	Visitor pre-register averaged 502 ms and was the slowest page.
10	Data integrity — records after the load	No after-test check of Reports, Registrations, or an existing user was saved.	Not executed	Do not mark Pass until an admin checks those pages in a browser.

Part D — Performance / Stress Indicators

No.	Test area	Actual result	Status	Remarks
1	Login page response, normal load	Average 175 ms (min 72, max 861), error 0.00%, 10 users.	Pass	Well under a 3,000 ms campus-use line.
2	Dashboard / portal response, normal load	GET /admin average 562 ms, error 0.00%. Mixed portal TOTAL average 171 ms, error 0.00%.	Pass	Times are acceptable. The saved HTML title is Sign In, so this measures the response that was returned, not a confirmed admin screen.
3	Throughput, concurrent users	Mixed portal run: 9.7 requests/sec across 115 samples, error 0.00%, 5 users.	Pass	Soak reached 38.6 requests/sec with error 0.00%.
4	RFID latency	Heartbeat average 123 ms (min 47, max 393), error 0.00%, 60 samples.	Pass	Under the 350 ms local target. The earlier scan averaged 186 ms but failed the token check.
5	Error rate under higher concurrency	30-user login: error 0.00%. 10-minute soak: error 0.00% on 23,189	Pass	No data-corruption check was saved, so this pass is for error rate

		samples.		only.
6	Stress limit	At 30 users the login average improved from 175 ms to 148 ms and error stayed 0.00%. A 50-user run was not performed.	Not executed	Degradation was not reached. The failure point is not known.

Part E — Reliability Evaluation Rating

Scale: 5 Strongly Agree, 4 Agree, 3 Neutral, 2 Disagree, 1 Strongly Disagree, N/A not run.

No.	Statement	Rating	Why this rating
1	Maturity — consistent during repeated runs	4 Agree	Login, registration, and the soak repeated with 0.00% error. The RFID scan was consistent too, but only as a token rejection until the token was corrected.
2	Availability — ISCVMS stayed accessible	5 Strongly Agree	Login, home, register, visitor pre-register, and the portal URLs kept returning through the 10-minute soak.
3	Fault tolerance — invalid requests did not take the site down	3 Neutral	A bad RFID token was rejected in JSON and the site stayed up. Wrong-password and malformed-body runs were not saved.
4	Recoverability — acceptable behavior after a stress peak	4 Agree	The 30-user run did not leave the site slow. The soak after the other runs stayed at 0.00% error. A heavier peak was not run.
5	Stability — soak did not become unusable	5 Strongly Agree	23,189 samples in 10 minutes, error 0.00%, max 2,793 ms.
6	Data integrity — sampled records stayed accurate	N/A	No post-test record check was saved.
7	Gate API reliability	4 Agree	Heartbeat: 60/60, avg 123 ms, ok true. The scan run was a token rejection, not an outage.
8	Portal reliability under concurrent load	3 Neutral	Five users, 0.00% HTTP error, avg 171 ms. The sampled response body is the Sign In page, so the admin screens themselves were not confirmed.
9	Error handling stayed clear and the system stayed stable	4 Agree	The RFID denial was a clear JSON message, and the next corrected heartbeat succeeded.
10	Overall reliability for a pilot	4 Agree	Public pages and the gate heartbeat are strong enough to cite. Confirm logged-in portals, re-run the scan with the valid token, and do the data check before calling every case Pass.

Part F — Open-Ended Questions

Which JMeter scenarios showed the most stable behavior?

The 10-minute soak and the RFID heartbeat. The soak handled 23,189 requests with 0.00% error. The heartbeat handled 60 requests with 0.00% error and an average of 123 ms. The 10-user and 30-user login-page runs were also stable, both at 0.00% error.

At what load level did the system start to degrade, and what failed first?

It did not degrade in the runs that were saved. Moving from 10 users to 30 users on GET /login lowered the average from 175 ms to 148 ms, and the error rate stayed 0.00%. The slowest sustained page was visitor pre-registration, at an average of 502 ms during the soak, with a maximum of 1,516 ms. A 50-user run was not performed, so a failure point was not found. The first hard failure in the set was the RFID scan, and that failure was a rejected API token, not a load collapse.

Were any reliability risks found that must be fixed before deployment?

No outage or data-corruption failure was observed. Three gaps should be closed before the results are treated as complete: the logged-in admin pages were not confirmed because the saved HTML is the Sign In page; the RFID scan must be repeated with the token the live server accepts; and wrong-password, malformed-request, and after-test record checks were not saved. The 5-login-per-minute limit was not what failed in these runs.

What hardware or network limits may have affected the JMeter results?

The laptop is an AMD Ryzen 5 7520U with 7.2 GB RAM, on a home internet connection to the public HTTPS site, not localhost. JMeter and the browser shared that PC. The live site also limits login to 5 posts per minute per email, registration to 3 posts per minute per IP, RFID scan to 30 per minute, and RFID heartbeat to 120 per minute. The scan run stayed inside the scan limit and still failed because the token was rejected.

Recommendations to improve ISCVMS reliability and performance

Keep the rate limits. Repeat the RFID scan after the JMeter header carries the live RFID_API_TOKEN, with no extra ${ } around the value, and expect a card-not-registered denial for TEST0001. Confirm that POST /login is followed by a real admin page before citing portal reliability. The visitor pre-register page was the slowest soak page and is the first public page to profile if a faster form is needed. Run any 50-user test off-peak, then check Reports and one existing user so data integrity is no longer N/A.

Acknowledgment

Apache JMeter 5.6.3 tests were performed on https://www.iscvms.com. The metrics above are the observed Summary Report values. Items marked Not executed were not saved and are not claimed as passes.

Tester / Evaluator: Adrian Adams M. Daduya

Date: 18 October 2026

Signature: _____